

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 49

INTERNATIONAL ENVIRONMENTAL CONVENTIONS - 2

INTERNATIONAL ENVIRONMENTAL CONVENTIONS

1 CONVENTION ON BIOLOGICAL DIVERSITY

■ The International Treaty on Plant Genetic Resources for Food and Agriculture

- ✓ Revision of the text of the International Understanding on Plant Genetic Resources was undertaken by **FAO** to harmonize its provisions with **CBD**, the outcome was **ITPGR**.
- ✓ ITPGR is popularly known as the '**Seed Treaty**'.
- ✓ The International Treaty on Plant Genetic Resources for Food and Agriculture was adopted by the **Thirty-First Session** of the Conference of the Food and Agriculture Organization of the United Nations on **3 November 2001**.



**Food and Agriculture
Organization of the
United Nations**

INTERNATIONAL ENVIRONMENTAL CONVENTIONS

1

CONVENTION ON BIOLOGICAL DIVERSITY

- The Treaty aims at
 - ✓ Recognizing the enormous contribution of farmers to the diversity of crops that feed the world;
 - ✓ Establishing a global system to provide farmers, plant breeders and scientists with access to plant genetic materials;
 - ✓ Ensuring that recipients share benefits they derive from the use of these genetic materials with the countries where they have been originated.



1 CONVENTION ON BIOLOGICAL DIVERSITY

MAIN PROVISIONS

1 MULTILATERAL SYSTEM

- The Treaty's solution to access and benefit sharing is the 'Multilateral System'.
- The system puts 64 of our most important crops – crops that together account for 80 percent of the food we derive from plants – into an easily accessible global pool of genetic resources that is freely available to potential users in the Treaty's ratifying nations for some uses.
- The Treaty facilitates access to the genetic materials of the 64 crops in the Multilateral System for research, breeding and training for food and agriculture.
- Those who access the materials must be from the Treaty's ratifying nations and they must agree to use the materials totally for research, breeding and training for food and agriculture.
- The Treaty prevents the recipients of genetic resources from claiming intellectual property rights over those resources in the form in which they received them.

1 CONVENTION ON BIOLOGICAL DIVERSITY

MAIN PROVISIONS

2 FARMER'S RIGHTS

- The Treaty recognizes the enormous contribution farmers have made to the ongoing development of the world's wealth of plant genetic resources.
- It calls for protecting the traditional knowledge of these farmers, increasing their participation in national decision-making processes and ensuring that they share in the benefits from the use of these resources.



THE UNITED NATIONS CONVENTION TO COMBAT DESERTIFICATION (UNCCD)

- Established in 1994, the UNCCD is the sole legally binding international agreement addressing the issues of Desertification, Land Degradation and Droughts (DLDD).
- The Convention addresses specifically the arid, semi-arid and dry sub-humid areas, known as the drylands, where some of the most vulnerable ecosystems and people can be found.
- UNCCD plays a central role in addressing these issues through capacity-building, the sharing of successful experiences, technology transfer, the provision of scientific support, awareness-raising, mobilization of resources and the provision of assistance to countries in implementing policies at national, subregional, regional and international levels.



United Nations

Convention to Combat
Desertification



THE UNITED NATIONS CONVENTION TO COMBAT DESERTIFICATION (UNCCD)

DRYLANDS

- Drylands are defined by a scarcity of water.
- Drylands can be classified into 4 types
 1. dry sub-humid lands
 2. semi-arid lands
 3. arid lands
 4. hyper-arid lands



DRY SUB-HUMID LAND



SEMI-ARID LAND



ARID LAND



HYPER ARID LAND

THE UNITED NATIONS CONVENTION TO COMBAT DESERTIFICATION (UNCCD)

DRYLANDS

- The UNCCD excludes hyper-arid zones from its definition of drylands.
- Drylands are zones where precipitation is balanced by evaporation from surfaces and by transpiration by plants (evapotranspiration).
 - ✓ Dry and sub-humid lands, including arid and semi-arid regions, grasslands, savannahs, and Mediterranean landscapes, encompass approximately 47% of the Earth's terrestrial area.
 - ✓ Dry and sub-humid lands include many fragile environments that need priority attention to avoid irreversible loss of biological diversity.
 - ✓ Wetland areas in dry and sub-humid lands, for instance, are often of crucial importance in supporting migratory bird species.



THE UNITED NATIONS CONVENTION TO COMBAT DESERTIFICATION (UNCCD)

DRYLANDS

- Dry and sub-humid lands also include important areas of extraordinary endemism, such as the Mediterranean Basin, which is home to more than 11,700 endemic plant species.
- The biodiversity of dry and sub-humid lands is well adapted to the harsh conditions - inconsistent rainfall patterns and high temperatures.
- The sociable weaver of southern Africa builds communal nests which can weigh up to 1,000 kg in order to maximize insulation from extreme temperatures.
- Many species of desert toads burrow into the sand and lie dormant for months until the return of the rains.



THE UNITED NATIONS CONVENTION TO COMBAT DESERTIFICATION (UNCCD)

IMPORTANCE OF DRYLANDS

- Drylands are home to more than 2.5 billion people.
- They harbor some of the world's most valuable and rarest biodiversity.
- They are major contributors to the world's breadbasket, considering that one in every three crops under cultivation today has its origins in the drylands.
- They are valuable indigenous food vaults because the wild ancestors and relatives of these plants still grow there.
- Drylands support 50% of the world's livestock and account for nearly half of all cultivated systems
- Drylands take up 41.3% of the land surface, and up to 44% of all the world's cultivated systems are in drylands.
- These lands are largely used as rangelands and currently support 50% of the world's livestock.
- Drylands are most vulnerable to DLDD, especially under the current climate change scenario.



THE UNITED NATIONS CONVENTION TO COMBAT DESERTIFICATION (UNCCD)

INTENSITY OF THE PROBLEM OF LAND DEGRADATION

- The scale of land degradation is immense, according to the report produced by the UNCCD in 2015, 10 to 20 percent of all land worldwide is degraded, amounting to between six and 12 million square kilometers.
- The consequences of land degradation are high, with widespread loss of biodiversity and ecosystem services.
- Annual global economic losses due to deforestation and land degradation were estimated at 1.5 - 3.4 trillion euros in 2010.
- Desertification is threatening the livelihoods of 1 billion people in over 100 countries and each year 12 million hectares of arable land is lost to drought.
- The extent and severity of land degradation worldwide, combined with the negative effects of climate change, population growth and an ever-increasing demand for natural resources, requires immediate action.



Avoiding land degradation through sustainable land management is the solution.

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